SE Lab 6

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Section: H

Initial run for code and the branch coverage for the initial test case:

Line coverage:

```
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote$ cd LAB6
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage run -m pytest test_phone_repair_quote.py
============================================================ test session starts =============================================================
platform linux -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6
collected 2 items

test_phone_repair_quote.py ..                                                                                                        [100%]

============================================================= 2 passed in 0.08s ==============================================================
```

```
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage report -m
Name                         Stmts   Miss  Cover   Missing
----------------------------------------------------------
phone_repair_quote.py           40     22    45%   6, 8, 11, 15, 17, 20, 26, 30-33, 38-50
test_phone_repair_quote.py       6      0   100%
----------------------------------------------------------
TOTAL                           46     22    52%
```

Branch coverage:

```
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage run --branch -m pytest test_phone_repair_quote.py
============================================================ test session starts =============================================================
platform linux -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6
collected 2 items

test_phone_repair_quote.py ..                                                                                                        [100%]

============================================================= 2 passed in 0.06s ==============================================================
```

```
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage report -m
Name                         Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------
phone_repair_quote.py           40     22     28      8    38%   6, 8, 11, 15, 17, 20, 26, 30-33, 38-50
test_phone_repair_quote.py       6      0      0      0   100%
-----------------------------------------------------------------------
TOTAL                           46     22     28      8    43%
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$
```

After adding the following test cases:

```
# repair_quote
def test_unknown_part():
    with pytest.raises(ValueError):
        repair_quote("speaker", "budget", False, False)


def test_unknown_brand_tier():
```

```
    with pytest.raises(ValueError):
        repair_quote("screen", "premium", False, False)


def test_flagship_screen():
    assert repair_quote("screen", "flagship", False, False) == 8600


def test_flagship_camera():
    assert repair_quote("camera", "flagship", False, False) == 5300


def test_budget_battery_no_warranty():
    assert repair_quote("battery", "budget", False, False) == 1800


def test_warranty_non_screen_is_free_parts():
    assert repair_quote("battery", "budget", True, False) == 300


def test_warranty_does_not_cover_screen():
    assert repair_quote("screen", "budget", True, False) == 4600


def test_express_surcharge():
    assert repair_quote("battery", "budget", False, True) == 2300


# repair_days
def test_no_technicians_in_stock():
    assert repair_days("battery", True, 0) == 3


def test_screen_out_of_stock():
    assert repair_days("screen", False, 1) == 6


def test_camera_out_of_stock():
    assert repair_days("camera", False, 1) == 6


def test_battery_out_of_stock():
    assert repair_days("battery", False, 1) == 3
```

```
def test_charging_port_out_of_stock_no_technicians():
    assert repair_days("charging_port", False, 0) == 5


# trade_in_value
def test_trade_in_invalid_price():
    with pytest.raises(ValueError):
        trade_in_value(0, 10, False)


def test_trade_in_negative_price():
    with pytest.raises(ValueError):
        trade_in_value(-100, 10, False)


def test_trade_in_negative_age():
    with pytest.raises(ValueError):
        trade_in_value(10000, -1, False)


def test_trade_in_under_12_months():
    assert trade_in_value(10000, 6, False) == 6000


def test_trade_in_exactly_12_months():
    assert trade_in_value(10000, 12, False) == 6000


def test_trade_in_13_to_24_months():
    assert trade_in_value(10000, 18, False) == 4000


def test_trade_in_exactly_24_months():
    assert trade_in_value(10000, 24, False) == 4000


def test_trade_in_over_24_months():
    assert trade_in_value(10000, 36, False) == 2000


def test_trade_in_exactly_48_months():
    assert trade_in_value(10000, 48, False) == 2000
```

```
def test_trade_in_over_48_months():
    assert trade_in_value(10000, 49, False) == 0


def test_trade_in_cracked_screen():
    assert trade_in_value(10000, 6, True) == 3000
```

```
amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote$ source venv/bin/activate
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote$ cd LAB6
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage run -m pytest test_phone_repair_quote.py
============================================================ test session starts ============================================================
platform linux -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6
collected 26 items

test_phone_repair_quote.py ..........................                                                                                  [100%]

============================================================ 26 passed in 0.60s =============================================================
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage report -m
Name                          Stmts   Miss  Cover   Missing
-----------------------------------------------------------
phone_repair_quote.py            40      0   100%
test_phone_repair_quote.py       59      0   100%
-----------------------------------------------------------
TOTAL                            99      0   100%
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage run --branch -m pytest test_phone_repair_quote.py
============================================================ test session starts ============================================================
platform linux -- Python 3.14.4, pytest-9.2.0, pluggy-1.7.0
rootdir: /mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6
collected 26 items

test_phone_repair_quote.py ..........................                                                                                  [100%]

============================================================ 26 passed in 0.18s =============================================================
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$ coverage report -m
Name                          Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------
phone_repair_quote.py            40      0     28      0   100%
test_phone_repair_quote.py       59      0      0      0   100%
-------------------------------------------------------------------------
TOTAL                            99      0     28      0   100%
(venv) amrita@Amrita:/mnt/c/Users/amrit/L6_L7_59_phone_repair_quote/LAB6$
```